You Can Just Do Things

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All statistics in this book reflect the blog as of November 27, 2025.

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You Can Just Do Things

A Data Portrait of John D. Cook's Blog

Owen D. Goode

2025

For John, For Everything

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Foreword

My freshman year of college, I was sitting in an honors ethics seminar at 8am on a Thursday. Douglas Weathers, a senior math major, walked in and made a beeline for me. “Last night I found out I follow your dad on Twitter! I love his stuff!” And that was when I realized that my dad’s blog was a bigger deal than any of us ever expected. That moment was in 2013, and the blog and its following have only grown since then.

I mentioned offhandedly at one point that it would be funny to collect some stats on the stats blog, and here we are—a deep analysis of the first seventeen years of my dad’s blogging career.

Through this exercise, I learned that way more people than Douglas Weathers care what my dad has to say in the world of statistics—some of these citations are highly impressive and I had no idea until we started looking them up. This compilation of anecdotes, equations, and general fun facts tied to statistics is a pillar of a niche community—as is the man who wrote it all. To me, John Cook is a humble, hardworking, loving father, but to this part of the Internet he is an eloquent, intellectual, creative savant. I’m grateful for the chance to understand more of this side of him, and look forward to putting more numbers to the scale of his success.

Thank you, Dad, for supporting me through calculus, job hunts, and parenting. And thank you, reader, for supporting my dad through the notebook, the crucible, and the institution. Here’s to the next seventeen years of blogging.

Elizabeth Goode (née Cook)

Preface: You Can Just Do Things

In the source code of pandas—the Python library used by millions of data scientists—there's a comment in `groupby.pyx`:

```
# Running stats update based on RunningStats::Push from
# https://www.johndcook.com/blog/skewness_kurtosis/
```

In OpenTSDB, the time-series database that powers monitoring at scale, there's a Java comment:

```
/**
 * This implementation is based upon a paper by John D. Cook,
 * which itself is based upon a method that goes back to a 1962
 * paper by B. P. Welford.
 */
```

In the official TensorFlow documentation, there's a note in the autodiff guide:

> Note: Don't actually invert the matrix.

These citations, buried in production code and embedded in official documentation, trace back to a WordPress blog that has been publishing since January 2008. One mathematician, writing about what interested him, day after day, for seventeen years.

This book is about what that looks like.

Simple numbers: 5,233 posts. 6,533 days. An average of one post every 1.25 days. No breaks longer than eleven days. No sabbaticals, no pivots to video, no platform migrations. Just writing.

But numbers don't capture influence. Cook's gamma function implementation has been ported to ten programming languages. His explanation of "don't invert that matrix" became the canonical reference that TensorFlow links to. His running statistics algorithm powers skewness and kurtosis calculations in pandas groupby operations—meaning data scientists computing higher moments on grouped DataFrames are running code that traces back to his blog.

This influence is infrastructure, not fame or brand.

Why does this matter?

Because it demonstrates something rare: the ability to just do things. Write about what interests you. Publish without permission. Build utility over time. Become genuinely useful without virality, platform, or strategy.

Cook's blog appears in university syllabi at MIT, Stanford, and Carnegie Mellon. When a programmer searches "gamma function python" at 2 AM, trying to debug something urgent, they find his explanation. When a graduate student needs an intuitive explanation of Bayesian inference, they find his post. When an engineer needs to understand why naive variance computation is numerically unstable, they find his running statistics tutorial. And then that algorithm ends up in their production code.

Some reach can be measured: 5,881 citations on Google Scholar. Stack Overflow answers that have reached 5.7 million users. Over 1,200 submissions to Hacker News, accumulating 25,000 upvotes. Seventeen specialized Twitter accounts with 500,000 combined followers. A single interpolation calculator page—one of more than eighty standalone tools—has generated over 600,000 search clicks in sixteen months.

Real influence, though, can't be counted. It lives in code comments, in understanding that went uncited, in problems solved at 2 AM that never got recorded.

This is not a biography—Cook's personal life isn't the subject. It is a data portrait of sustained intellectual work, an attempt to see the shape of that influence through counting, measuring, and visualizing. How many posts about prime numbers? When did cryptography become a major topic? How has the average post length changed?

Data reveals patterns. Patterns suggest a story. An arc emerges from scattered consultant's notebook to mathematical institution: from short posts about PowerShell bugs in 2008 to deep explorations of elliptic curve cryptography in 2025.

It asks what happens when someone just keeps going.

When the blog launched in January 2008, people had blogrolls. Google Reader existed. Twitter asked "what are you doing?" instead of optimizing for engagement.

While the internet reinvented itself three times over, johndcook.com kept doing one thing: math, computing, statistics, the occasional digression into music theory or typography. No ads. No viral moments. No controversy.

Ecology has a term for this: climax species—the organism that remains after initial colonizers have come and gone, after boom-and-bust cycles have played out, after the ecosystem has stabilized.

Viral sensations aren't the internet's climax species. Quiet sites that just keep being useful are.

Consider the comparison. Joel Spolsky's *Joel on Software* shaped a generation of developers, then he co-founded Stack Overflow and largely stopped blogging. Jeff Atwood's *Coding Horror* ran

for years before the same pivot. Steve Yegge wrote legendary posts at Google, then went quiet. Paul Graham's essays remain influential but arrive sporadically. These are the famous names in technical blogging—and none of them kept going the way Cook has. They built platforms, started companies, moved on. Cook just kept writing. Seventeen years, same site, same format, same focus on being useful. The rarity isn't the quality. It's the duration.

Seventeen years in three acts.

First, the beginning: 2008 to 2012, when the blog was a consultant's notebook, scattered, practical, exploratory. Second, the crucible: 2013 to 2017, when practice was tested and voice clarified. Third, the institution: 2018 to 2025, when the blog became infrastructure, cited in documentation, ported across languages, running in production systems worldwide.

Data tells this story. Data also misses things. A final chapter acknowledges what counting can't capture: the reader who finally understood, the code that was never attributed, the influence that left no trace.

This book is a map. The territory is larger.

All statistics in this book reflect the blog as of November 27, 2025. The archive continues to grow.

Chapter 1

The Shape of the Start

"Moore's law and software bloat"
Title of the first post, January 9, 2008
(182 words)

On January 9, 2008, the first post appeared. It observed that software expands to consume whatever hardware speed Moore's law provides. One hundred eighty-two words. No code. No equations. Just a small observation.

A second post the same day: "Integration surprises." One hundred thirty-four words about the practical value of techniques that seemed impractical in college.

Cook posted twice on his first day. He posted again on January 10. And 11. And 12. By the end of January, there were 37 posts. By the end of 2008, there were 410, capturing the energy of beginning.

The Launch Year

2008 stands alone in the archive. Four hundred and ten posts in a single year, more than one per day on average. The posts are short: 206 words on average, some under 100. They read like notes jotted between other work.

A quick observation about software bloat. A pointer to an interesting integral. A link to someone else's insight. A tip about PowerShell. A note on clinical trial methodology.

The topics scatter across a consultant's working life:

Category	Posts in 2008
Software development	86
Math	85
Statistics	81
Computing	35
Business	29

In 2008, this wasn't a math blog—it was a consultant's notebook, a professional sharing whatever seemed useful that day.

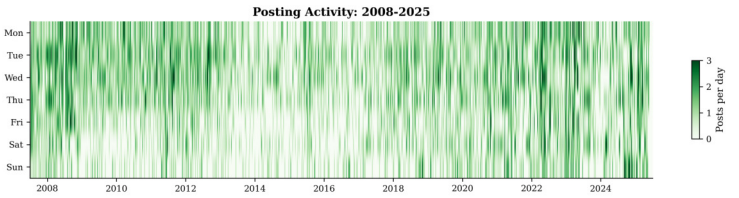


Figure 1.1: Activity heatmap, 2008-2025

The First Posts

What a working consultant found interesting in 2008:

January 9: "Moore's law and software bloat": Software expands to fill available hardware.

January 11: "Million dollar cutoff for software technique": A practical threshold for when formal methods pay off.

January 14: "Tips for learning regular expressions": The tools of the trade.

January 21: "Selection bias and bombers": A curiosity, a calculation, a small insight.

These were not the posts of someone building a platform, but of someone thinking out loud—a helpful but tentative voice making no claim to authority.

Worth quoting the very first post in full:

> I ran across an article recently comparing the performance of a 1986 Mac and a 2007 PC. Of course the new machine would totally blow away the old one on a number crunching benchmark, but when it comes to the most mundane benchmarks — time to boot, launch Microsoft Word, open a file, do a search and replace, etc. — the old Mac pulls ahead slightly. Software bloat has increased at roughly the same rate as Moore's law, making a new machine with new software no better than an old machine with old software in some respects.
>
> The comparisons in the article resonate with my experience. I expect administrative tasks to be quick and number crunching to be slow, and so I'm continually surprised how long routine tasks take and how quickly numerical software runs.
>
> Update (December 2014): Processors have not gotten much faster in the time since this post was first published, though we have more of them. However, applications do open faster. Perhaps software vendors have paid attention to bloat precisely because the free lunch of faster processors is over.

One hundred eighty-two words that started a seventeen-year practice.

The Consultant's Breadth

In 2008, Cook was a research statistician at MD Anderson Cancer Center, consulting on clinical trial design. But he'd also been a software developer, a project manager, and an independent consultant—someone who worked on many things, with breadth reflecting that history.

Clinical trials appear in 55 posts over the blog's lifetime, concentrated in the early years. By 2018, they had faded to a single post. The topic was professional obligation, not personal fascination.

PowerShell appears in 47 posts: the tool of a consultant working in Windows environments. Cook wrote about PowerShell from 2008 to 2012, with the heaviest concentration in the first two years. Then PowerShell fades, replaced by Python.

Perl gets 20 posts scattered across 2008-2021, with the last appearing in March 2021.

These are fossils, preserving what a working consultant needed to know in the late 2000s. Their fading reveals temporary interest versus permanent fascination.

The Early Rhythm

Even in 2008, patterns emerge.

Tuesday leads with 86 posts. **Wednesday** and **Thursday** follow with 65 each. Weekends trail: Saturday has 48, Sunday just 35.

This is a working professional's rhythm—attention when work allows, weekday mornings, before the demands take over. Already visible in year one: the 7 AM peak that persists through all seventeen years.

Morning dominates. In 2008, the hour from 7:00-7:59 AM accounts for a little over 10% of posts; across the full corpus it reaches 13%. Add the hours from 6-9 AM and you have about 30% of 2008's output and 43% of all posts over seventeen years.

Cook was writing before work. Or perhaps writing *was* work: a way to warm up the mind, to think through problems, to share what seemed interesting before the meetings started.

The Short-Post Era

The median post length in 2008 was 163 words—half of all posts were shorter than that.

Some were mere announcements: "New prime number record" (9 words). Others were pointers: "Technology hype curve" (12 words). The blog's **median** post length overall is 312 words, but in 2008 it was just 163. The blog served a curatorial function in 2008, directing attention more than providing explanation.

The deep dives would come later. First came the practice establishing itself—the habit of showing up, of publishing something, anything, every day.

Quality would follow quantity, but the habit of quantity had to be established first.

What 2008 Reveals

What beginning looks like:

Scattered. Topics ranging from software bloat to Chinese Unicode to clinical trials to PowerShell. No focus because focus hasn't been found yet.

Short. Two hundred six words on average: enough to make a point, not enough to develop an argument. Notes, not essays.

Frequent. More than one post per day. Beginning's energy, before sustainability matters.

Professional. Topics reflecting a consultant's working life. Serving the job.

Many new blogs start this way. Few sustain it. Typical pattern: a burst of enthusiasm, then gradual abandonment. The internet littered with blogs that have three posts from 2009 and then nothing.

Cook's first year was not a burst that would fade—it was a beginning.

The Years That Followed

2008 established the habit. Following years tested it.

Year	Posts	Avg Words
2008	410	206
2009	366	256
2010	284	268
2011	317	222
2012	357	216

Post count declined from 410 to 284-366, then stabilized. Average length crept up from 206 to 256-268. The trade-off between quantity and depth was beginning: more words per post, fewer posts per year.

But the habit held. Three hundred or more posts every year. Practice taking root.

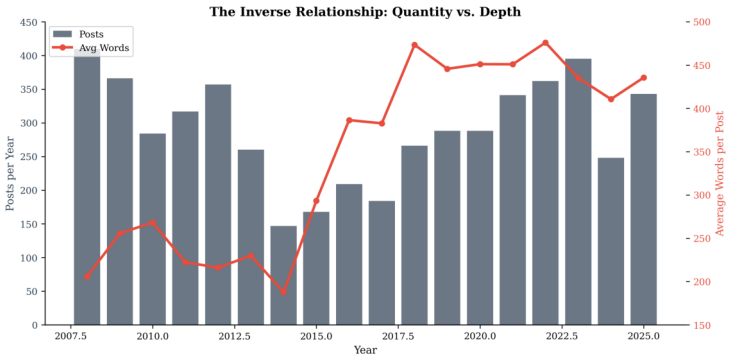


Figure 1.2: Posts and length by year

The Notebook Phase

Years 2008-2012 form the first era. Call it the Notebook Phase.

Total posts: 1,734 Average length: 232 words Dominant categories: Software development, Math, Computing, Business Characteristic posts: Tips, pointers, short observations

A consultant's scratchpad. Useful observations from working life. Tools and techniques. Quick notes on mathematical curiosities. Nothing claiming to be comprehensive or authoritative.

The voice was helpful but not yet distinctive, the topics reflected professional needs, and the posts served the moment rather than building toward permanence.

Remarkable not for quality but consistency. Five years of showing up. Five years of writing something, even when there was nothing profound to say. Five years building the habit that would sustain what came next.

Start before you're ready. The 2008 blog wasn't waiting for quality; it was generating quantity until quality emerged. Four hundred ten posts of variable merit, but four hundred ten chances to find what worked.

Chapter 2

The Consultant's Toolkit

The blog has categories. WordPress requires it: you cannot publish a post without assigning it to at least one category. Many bloggers treat this as a chore, clicking the obvious option and moving on.

Cook has twenty-six categories. In the early years, they reveal a consultant's working life.

The Professional Landscape

The category breakdown from 2008-2012 tells a story of someone solving problems across domains:

Category	Posts (2008-2012)	% of Era
Uncategorized	388	22%
Math	357	21%
Software development	327	19%
Statistics	205	12%
Computing	145	8%
Business	126	7%

The spread is remarkably even. Math, software development, and uncategorized posts each claim around 20%. Statistics, computing,

and business together account for another quarter. Not yet a math blog, but a professional's blog that includes math.

The category "Software development" contains posts like "Tips for learning regular expressions" and "Literate programming and statistics." The category "Business" contains posts like "Task switching" and "Organizational scar tissue."

These are the concerns of someone who works in organizations.

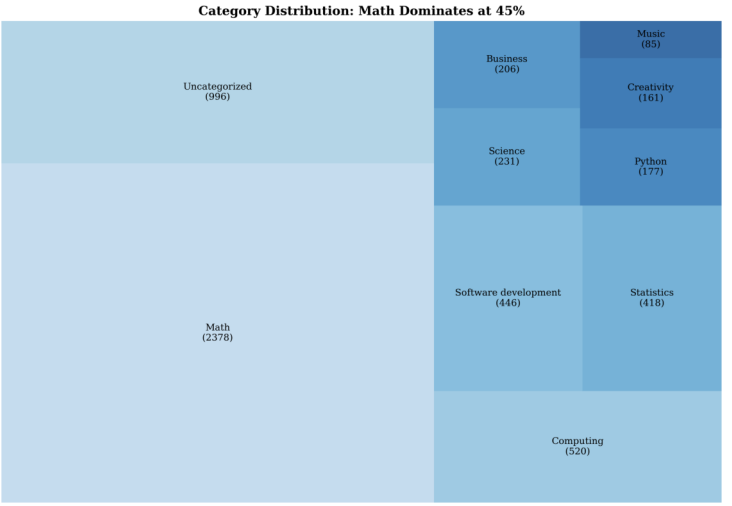


Figure 2.1: Category distribution

The Tools of 2008

The early archive preserves a snapshot of the consultant's toolkit.

PowerShell gets its own category (47 posts total). Cook wrote about PowerShell extensively from 2008-2012, with the heaviest concentration in the first two years. In February 2009, he linked to Joe Pruitt's "PowerShell A-Z" series. This was the tool of someone automating tasks in Windows environments. The last PowerShell post appeared in August 2012.

Perl appears in 20 posts across the blog's lifetime. "Periodic table of Perl operators" (2008). "Perl One-Liners Explained" (2012). "Learn one Perl command" (2012). Practical tools for practical problems. The last Perl post appeared in March 2021: "tcgrep: grep rewritten in Perl."

Mathematica gets 74 tag mentions across all years. It's a tool Cook has used consistently, though it never dominated. The early posts show someone who reached for whatever tool fit the problem.

The language distribution in the early era:

Language	Posts (2008-2012)
Python	77
PowerShell	48
C++	33
Perl	11
Mathematica	9

Python already leads in the early years, though it's far from dominant. By 2025, it would grow to 207 tagged posts.

The Clinical Trials Domain

Fifty-five posts carry the "Clinical trials" category. They cluster in 2008-2013, then fade. By 2018, just one remained.

This was Cook's day job. From 2000 to 2013, he worked as a research statistician at MD Anderson Cancer Center, designing clinical trials for cancer treatments. He earned his PhD in mathematics from UT Austin in 1992, did a postdoc at Vanderbilt, and spent 1995-2000 in software development and project management before the MD Anderson role. The blog posts from this era reflect that work:

"Probability that a study result is true" (2008) "Bayesian clinical trials in one zip code" (2009) "Why drugs often list headache as a side-effect" (2010)

These posts were professional output—sharing methodology with colleagues, thinking through problems in public. They're technically dense. They assume an audience of statisticians working on similar problems.

When Cook left MD Anderson in 2013 to consult full-time, the clinical trials posts didn't follow. The topic was attached to a job, not to personal fascination.

The disappearance of clinical trials is evidence of what the blog would become: a place for what Cook chose to think about, not what he was paid to think about.

The Business of Consulting

The early blog contains something the later blog lacks: business advice.

206 posts carry the "Business" category. Most appear before 2015. Titles include:

"How to avoid being outsourced or open sourced" (2008) "Why programmers cannot be managed" (2008) "Medieval software project management" (2008) "Would you rather serve a market or a boss?" (2012)

These read like lessons learned. A consultant who has made mistakes and wants to save others the trouble. The voice is practical, not theoretical.

By the recent era, business posts nearly disappear. The transformation is complete: from consultant sharing professional wisdom to mathematician sharing mathematical fascination.

The Music Interlude

Eighty-five posts carry the "Music" category. They span all seventeen years, but they never dominate.

Music posts reveal the blog's range:

"Circle of fifths and number theory" (2009): the connection between musical intervals and modular arithmetic.

"Typesetting music with LilyPond" (2009): the tools for creating sheet music programmatically.

"The acoustics of Hagia Sophia" (2018): the physics of sound in architectural spaces.

Not the posts of a professional musician. The posts of a mathematician who notices that music is also math. The same curiosity that drives the special functions posts: what is the structure underneath the surface?

The music category demonstrates that even in the early era, the blog followed fascination, not just profession. Music was never a job. It was always a choice.

The Geography of Curiosity

Thirty posts carry the "Geography" category. Forty-two carry the "Orbital mechanics" tag.

These are the outliers: topics that appear, get explored, and persist at low frequency. They reveal a mind that finds mathematics everywhere:

"Finding distances using latitude and longitude" (2008) "Mercator projection" (2009) "Journey away from the center of the Earth" (2011)

The geography posts are typically short, practical, and include code. Someone needed to calculate something about the Earth, and Cook explained how.

Orbital mechanics posts similarly bridge abstract math and concrete application:

"Kepler and the contraction mapping theorem" (2018) "Orbital resonance in Neptune's moons" (2019) "Hohmann transfer orbit" (2020)

These posts demonstrate what the later blog would crystallize: mathematics as the universal tool, applicable to any domain that has structure.

What the Early Categories Reveal

The 2008-2012 category distribution shows a generalist. Someone who works on many things, writes about all of them, and hasn't yet discovered what will persist.

Compare to the 2020-2025 distribution:

Category	Posts (2020-2025)	% of Era
Math	1,274	64%
Uncategorized	305	15%
Computing	245	12%
Statistics	96	5%
Python	22	1%

Math has nearly doubled its share. Business has collapsed. Clinical trials has vanished. The consultant's toolkit has been replaced by the mathematician's focus.

The early era's breadth wasn't a mistake. It was exploration. The later era's focus emerged from that exploration—from five years of writing about everything, and discovering what kept being interesting.

The Uncategorized Mystery

Nine hundred and ninety-six posts carry the "Uncategorized" label.

Some of these are genuinely miscellaneous: posts that don't fit anywhere. But many could have been categorized and weren't. The Uncategorized bin grows over time, suggesting that as the blog matured, the category system became less important.

In the early era, categories helped organize a scattered practice. In the later era, the practice had its own coherence. The categories became formality.

Chapter 3

Finding the Rhythm

> "I've never had a schedule for this blog: I write when I feel like writing, which usually means several times a week."
>
> "Blogging pace" (2024)

Temporal patterns are autobiography written in timestamps. Every post carries a date and time. Aggregate 5,233 of them and a portrait appears: when one person does his thinking.

The Tuesday Peak

Tuesday leads most years, though Wednesday takes the top spot in several.

Day	Posts	% of Total
Tuesday	1,003	19.2%
Wednesday	938	17.9%
Monday	887	17.0%
Thursday	823	15.7%
Friday	561	10.7%
Saturday	555	10.6%
Sunday	466	8.9%

The pattern is stable. Tuesday and Wednesday compete for the top spot across years, with Tuesday winning overall. The weekend always trails. Not a hobby blogger working when inspiration strikes on lazy Sundays. A professional integrating writing into a workweek.

Why Tuesday? Perhaps Monday is for catching up on email and meetings. By Tuesday, the week has settled. There's time to write.

The Tuesday pattern survived career changes, platform changes, and seventeen years of life. It's an invariant: something about how Cook works that transcends context.

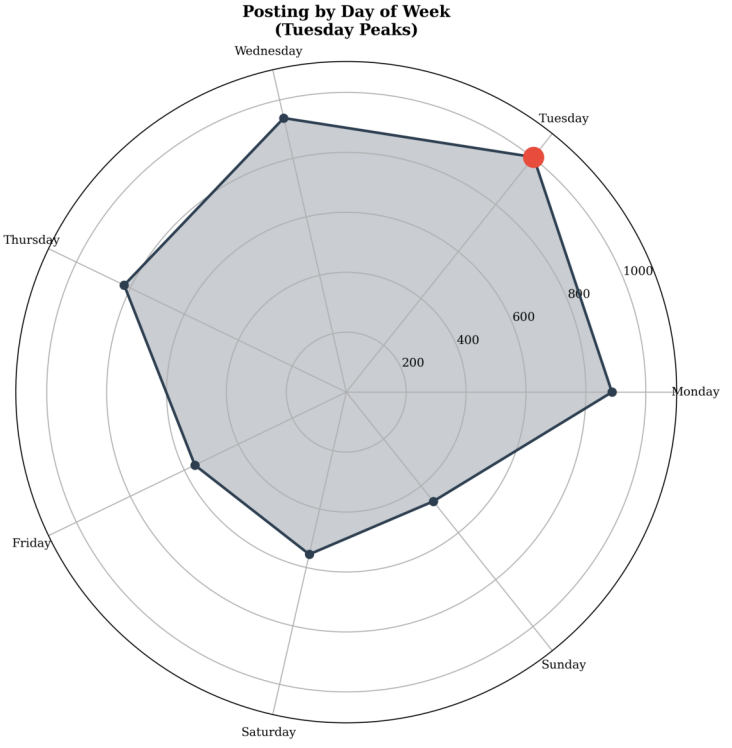


Figure 3.1: Weekly rhythm

The 7 AM Hour

The peak hour is 7:00 to 7:59 AM.

Hour	Posts	% of Total
7 AM	683	13.1%
8 AM	580	11.1%
6 AM	501	9.6%
9 AM	503	9.6%

Forty-three percent of all posts appear between 6 AM and 10 AM. The blog is a morning practice.

The pattern suggests writing before the day's demands take over. Before meetings, before email, before the work that doesn't leave room for reflection. The 7 AM post is the first intellectual act of the day.

Compare the morning cluster (6-10 AM: 2,267 posts) to the afternoon cluster (12-5 PM: 1,226 posts) to the evening cluster (6-10 PM: 770 posts). Morning wins by nearly two to one.

The Habit Before the Content

This rhythm existed from the beginning. The 7 AM peak is visible in 2008. The Tuesday lead is visible in 2008. The weekend slump is visible in 2008.

The habit preceded the content. Before Cook knew what the blog would become, he knew when he would write it.

> "The hard part isn't having ideas. The hard part is sitting down at the same time every day." — (Editorial observation)

This may be the secret to seventeen years. The rhythm was established early and protected fiercely. The content could change, from PowerShell to Python, from clinical trials to cryptography, but the rhythm stayed constant. Tuesday mornings. 7 AM. Before the rest of the day.

Monthly Flatness

Unlike days and hours, months show no pattern.

Month	Posts	%
January	469	9.0%
February	444	8.5%
March	436	8.3%
April	406	7.8%
May	394	7.5%
June	413	7.9%
July	410	7.8%
August	438	8.4%
September	448	8.6%
October	480	9.2%
November	478	9.1%
December	417	8.0%

Each month accounts for 7.5-9.2% of posts. No summer slump. No December hiatus. No April surge.

This flatness is the signature of consistency. Seasonal variation would suggest a practice that competes with other life rhythms: vacations, holidays, academic calendars. The flatness suggests a practice that has become routine, independent of external calendars.

The Gaps

Every blogger takes breaks. The question is how long.

The longest gap in the entire archive is 11 days: July 9 to July 21, 2014.

The second-longest gaps are 8 days, appearing three times.

No gap reaches two weeks. No sabbatical. No "I'm back after six months." In seventeen years, the blog has never gone silent for more than eleven days.

Most professionals take longer vacations than the blog's longest gap. A two-week holiday would set a new record. A month away would be unprecedented. Whatever else was happening in Cook's life—travel, illness, family—the blog kept appearing.

What Rhythm Reveals

The temporal patterns reveal a practice designed to survive.

Morning writing gives the blog the best hours—before fatigue and distraction.

Weekday dominance integrates the practice with professional life rather than competing with it.

Monthly flatness means the blog doesn't depend on mood or season.

Short gaps prevent restart friction from accumulating.

Each property makes the practice more sustainable. Morning writing prevents falling behind. Weekday integration avoids the choice between work and blog. Monthly consistency avoids rebuilding momentum.

The rhythm is engineered for persistence. It's not the rhythm of inspiration. It's the rhythm of showing up.

The Practice as Foundation

The habit established in 2008-2012 was not just about writing. It was about when to write.

The rhythm that emerged in the first year—Tuesday mornings, 7 AM, consistent across months—became the foundation for everything that followed. When the content changed, the rhythm remained. When the voice evolved, the rhythm remained. When interests shifted, the rhythm remained.

This is what sustained practice looks like underneath: not just what you do, but when you do it, protected across years.

So What?

The rhythm was established before the content clarified. The sustainability came from engineering the practice around life's constraints—mornings before work, weekdays that fit professional rhythms, gaps short enough to prevent restart friction—not from waiting for inspiration. Seventeen years later, the content had transformed completely, but the Tuesday morning habit remained.

Chapter 4

The Quiet Year

Every long-running system has a moment where it almost stops.

For a startup, it's the "trough of sorrow." For a novelist, it's the middle of the manuscript where the initial energy fades and the ending is not yet in sight. For the blog at johndcook.com, that moment was 2014.

If you look at the timeline of the archive, 2008 to 2013 looks like a wall of noise. Four hundred posts. Three hundred posts. The output is relentless, a daily stream of notes, links, and observations. The blog was a consultant's notebook, filling up page after page with the sheer energy of beginning.

Then comes 2014.

The numbers drop off a cliff. One hundred and forty-seven posts.

It is the lowest point in the seventeen-year history. It is less than half the volume of the year before. Compared to the frenzy of the launch year, the blog had gone nearly silent.

To an outside observer, this looks like death. In the lifecycle of online writing, this is usually the end. The initial enthusiasm burns out. The professional obligations pile up. The writer misses a week, then a month, and the silence hardens into permanence. The internet is littered with blogs that have three hundred posts from 2009 and nothing since.

But 2014 wasn't an end. It was a filter.

The Eleven Days

The silence has a shape.

In a practice defined by consistency—Tuesday mornings, 7 AM, year after year—the gaps tell their own story. The longest gap in the entire seventeen-year archive occurred in this quiet year: July 9 to July 21, 2014.

Eleven days.

It sounds trivial. Most people take two-week vacations every year. Most writers disappear for months. But in the context of this specific machine, eleven days is a chasm. It is the only time the blog ever blinked.

We don't need to know the biographical details of those eleven days. We don't need to know if it was a vacation, a family crisis, or simply a moment of exhaustion. The data tells us what matters: the gap opened, and then it closed.

On July 21, the blog returned. The rhythm resumed. The machine restarted.

The significance of 2014 isn't that the blog almost stopped. It's that it started again.

If the blog had died in 2014, the pandas citation would never have happened. The TensorFlow documentation would link elsewhere. The gamma function would have been ported from someone else's code, or not at all. The ten-language implementation trail wouldn't exist. A developer in Mumbai, searching at 2 AM for a numerically stable variance algorithm, would have found a Stack Overflow answer instead of the primary source. The infrastructure would have been built differently, or not built at all.

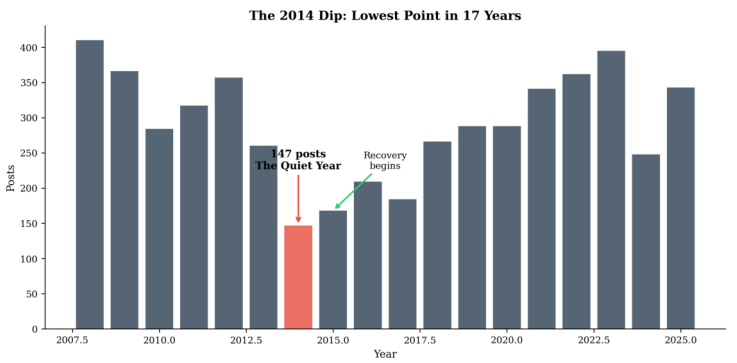


Figure 4.1: The 2014 dip: 147 posts, the lowest annual output in the blog's history. The recovery began in 2015.

The Pivot to Depth

If 2014 was a crisis of volume, it was also the birth of depth.

While the post count collapsed, something else was happening in the background. The "notebook" era, defined by short, quick pointers, was ending. The "essay" era was beginning.

Hidden inside the quiet year is one of the most substantial pieces of writing Cook ever produced. On February 24, 2014, he published "Writing down an unwritten language."

It is 1,601 words long. It is not a quick tip about PowerShell or a link to a math paper. It is a thoughtful, lyrical exploration of how missionaries create orthographies for spoken languages. It explores phonology, the ethics of transcription, and the mathematical problem of mapping sounds to symbols.

> "A writing system is an abstraction, a model of a language. Like all models, it highlights some features and suppresses others. It is a trade-off between accuracy and simplicity."

This is not a consultant trying to get clients. This is a writer wrestling with a complex idea.

In 2008, the average post was 206 words. "Writing down an unwritten language" was eight times that length. The silence of 2014 wasn't emptiness; it was concentration. The energy that used to go into publishing *every day* was being redirected into publishing *better things*.

What Burned Away

2014 burned away the noise.

The "Business" category began to fade. The "Clinical Trials" posts—vestiges of a former career—stopped entirely. The links to other people's articles became less frequent. The obligation to "feed the beast" with daily content was abandoned.

What remained was the core: Mathematics. Computing. The intersection of the two.

The 147 posts of 2014 are different from the 410 posts of 2008. They are less reactive. They are less about "what is happening in the industry right now" and more about "what is true forever."

The blog survived 2014 because it changed its function. It stopped being a log of a consultant's activity and became a library of a mathematician's interests. A log requires constant updating; if you stop logging, the log is dead. A library grows slowly. It doesn't need to shout. It just needs to accumulate.

The Lesson of the Dip

The lesson of the quiet year is that consistency doesn't mean uniformity.

A seventeen-year practice will not look the same in year seven as it did in year one. There will be winters. There will be droughts. There will be years where the work feels heavy and the output drops.

The mistake is to treat the dip as a failure. If Cook had looked at his stats in December 2014, he would have seen a line crashing downward. He might have concluded that he was "losing it," that the blog was dying, that he should give up.

But the data shows the opposite. 2014 wasn't the end. It was the foundation for the "Institution" phase that followed. The volume would return—by 2018, he was back to 266 posts a year—but the depth gained in the quiet year would stay.

The practice contracted so it could expand. It went quiet so it could find its voice.

So What?

Crisis forces clarification. The year that dropped output to 147 posts burned away business advice, clinical trial obligations, and the compulsion to feed daily content—leaving only what mattered. The habits that survived that pressure were the ones worth keeping: the Tuesday mornings remained, but they produced essays instead of links, libraries instead of logs.

Chapter 5

The Crucible

> "Number theory has traditionally been the purest of pure mathematics. People study number theory for the joy of doing so, not to make money. At least that was largely true until the advent of public key cryptography."
>
> "Surprising applications of math" (2011)

The recovery from the quiet year didn't look like a return to the past. It looked like something new.

In 2015 and 2016, the volume began to climb back up. But the shape of the posts had changed. The consultant's notebook—scrappy, reactive, brief—was gone. In its place was a practice of sustained explanation.

The years 2013-2017 form the crucible phase: a period when the blog was tested, refined, and transformed. What emerged was fundamentally different from what entered.

The Shift in Scale

In the "Notebook" era (2008-2012), the average post was 232 words. In the "Institution" era (2018-2025), the average post was 448 words.

The blog literally doubled in density.

The shift happened here, in the years following the 2014 dip. It wasn't a sudden jump; it was a gradual deepening.

Year	Average Words
2014	188
2015	293
2016	386
2017	383

By 2016, the new standard was set. A typical post was no longer a paragraph; it was an essay. It was substantial enough to contain code, a derivation, and a conclusion.

This change broke the "blogger's trade-off." The conventional wisdom says you can have high frequency or high depth, but not both. If you write every day, you write fluff. If you write essays, you publish once a month.

Cook found a way to do both. By 2018, he was publishing 266 times a year—nearly every weekday—and maintaining that 450-word average. He wasn't choosing between quantity and quality. He was using quantity to *refine* quality.

The Gamma Function Test

We can see the deepening by looking at how Cook treats a single topic over time.

Consider the Gamma function, $\Gamma(z)$. It's a generalization of the factorial function to complex numbers. It's notoriously tricky to implement numerically because it grows so fast—$\Gamma(100)$ has 156 digits.

Cook wrote about Gamma in the early years. In 2010, he posted "Diagram of gamma function identities." It was 114 words. It listed a few formulas, provided a link to a reference, and stopped. It was a note to self. *Useful*, but thin.

Compare that to the Gamma posts from the deepening era.

In 2017, he wrote "Approximate inverse of the gamma function" at 481 words. This wasn't a list of formulas. It was a full exploration. He explained the approximation method, provided Python code to implement it, and tested the accuracy across different ranges. He didn't just point to the math; he inhabited it. He translated it.

The pattern repeated across the gamma function posts of this era. Each one deeper than a reference manual, each one building working code, each one explaining not just *what* but *why*.

The 2010 post was a signpost: "This math exists." The 2017 posts were infrastructure: "Here is how to build with this math."

The End of the Linklog

The early internet was built on "linklogs," blogs that primarily pointed to other things. "Check this out." "Read this article."

In 2008, a significant portion of Cook's posts were essentially links. By 2017, the links were gone.

The blog stopped being a curator of other people's work and became a primary source. This is a critical threshold for any intellectual project. Are you filtering the stream, or are you adding to it?

The deepening meant Cook was no longer relying on external novelty to drive the blog. He was generating the novelty himself, from the inexhaustible mine of mathematics. You don't need to

link to the news when you have the Riemann Zeta function. It hasn't changed in 160 years, and it's still more interesting than today's tech headlines.

The Clinical Trials Extinction

Focus requires shedding. As the blog deepened into mathematics, it left things behind.

Year	Clinical Trials Posts
2008	24
2009	6
2010	6
2011	2
2012	6
2013	1
2014	1
2015	5
2016	3
2018	1

The pattern is extinction. Two dozen posts in 2008, then a steady decline to single digits, then effectively zero in recent years.

Clinical trials was Cook's day job at MD Anderson, where he spent thirteen years—first as a biostatistician designing trials, later in software development management. By the end, he was managing developers rather than doing mathematics. When he left in 2013 to found Singular Value Consulting, the topic began to fade from the blog. By 2018, it was down to a single post.

He has 5,881 Google Scholar citations, mostly from that clinical trials work. His paper on "dose-finding based on efficacy-toxicity trade-offs" remains influential in biostatistics. He could have built an audience around clinical trial methodology.

He didn't. The topic faded because it was professional obligation, not personal fascination. The business posts tell a similar story: 126 posts in the early years, many reflecting the frustrations of corporate software management, dwindling to 27 by the recent era. Going independent meant he could finally write about what he wanted to think about.

The PowerShell Fossil Record

PowerShell tells a similar story.

Cook wrote about PowerShell consistently from 2008-2012, producing 47 posts on scripting techniques, language features, and Windows automation. This represents serious investment in mastering the tool.

After 2012, PowerShell posts decline. The last PowerShell post appears in July 2014. The tool had been replaced.

The replacement was Python. The first SciPy-tagged post appeared in February 2009. By 2025, Python had accumulated 207 tagged posts. The transition was generational: Windows automation gave way to scientific computing.

What Rose as Others Fell

The shedding wasn't just loss. It created space.

Category	2008-2012	2018-2025	Change
Math	357 (21%)	1,611 (64%)	+43 pts
Statistics	205 (12%)	120 (5%)	-7 pts
Computing	145 (8%)	314 (12%)	+4 pts
Business	126 (7%)	27 (1%)	-6 pts

Clinical trials	44 (3%)	1 (0%)	-3 pts

Math went from about one-fifth to nearly two-thirds of output. The narrowing was dramatic.

This isn't accidental. Every hour spent on business advice was an hour not spent on number theory. Every PowerShell post was a Python post not written. The later blog's mathematical depth required abandoning breadth.

Quality Through Repetition

How do you double your output while doubling your depth?

The answer lies in the subject matter. Mathematics rewards repetition.

If you write about politics, yesterday's post is trash today. If you write about software frameworks, your tutorial from 2015 is broken by 2017.

But if you write about probability distributions, the work compounds. A post about the Beta distribution written in 2015 helps you write a post about the Dirichlet distribution in 2016. The code you wrote for numerical integration in 2014 becomes the library you import in 2017.

The deepening was possible because the blog found its true subject. Cook moved away from the ephemeral (tech news, business advice) and anchored himself in the permanent.

He wasn't just writing more. He was building a cathedral, brick by brick. The foundation laid in the quiet year supported the weight of the structure to come.

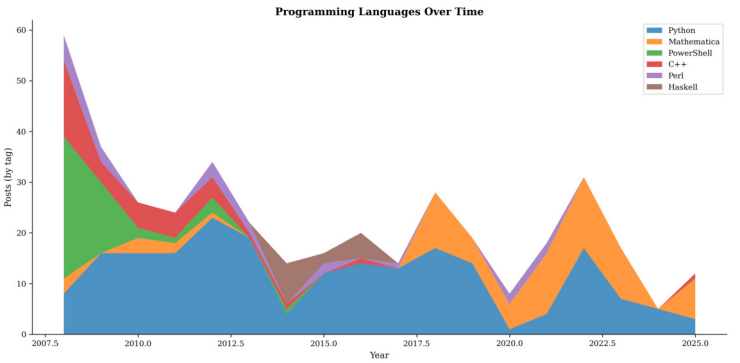


Figure 5.1: The language transition: PowerShell fades as Python rises. By 2018, Python dominates the blog's code examples.

What Persisted

Some things didn't change:

Pi (153 posts) spans all seventeen years. The first pi post appeared in 2008; the most recent in 2025.

Primes (103 posts) similarly persist. The mathematical core remained stable while everything around it shifted.

The Fourier transform (45 tagged posts) appears throughout. Signal processing was never a professional obligation; it was genuine interest.

Special functions (140 posts) never fade. The gamma function, Bessel functions, hypergeometric functions—these define the blog's mathematical identity.

The pattern: professional topics faded. Personal fascinations persisted. The blog became a map of what Cook actually wanted to think about, freed from what he was paid to think about.

So What?

The crucible phase refined the blog into its final form. Posts doubled in depth while professional obligations burned away. Clinical trials went to zero. PowerShell became a fossil. Business advice faded. What emerged was a blog about mathematics—not because the other topics failed, but because the crucible revealed what Cook actually wanted to think about when nobody was paying him to think. The transition from 21% mathematics to 64% mathematics wasn't narrowing. It was clarification.

Chapter 6

The Mathematical Core

> "The thing I enjoy most about math is seeing surprising connections."
>
> John D. Cook

In the beginning, the blog was a mixture. Math, code, business, statistics.

By 2018, the mixture had separated. The volatile elements—business advice, quick tips, tech news—had evaporated. What remained was the solute.

The blog became a mathematics blog.

The Dominance of Math

The shift is visible in the category data.

Era	Math Posts	% of Total
2008-2012	357	21%
2013-2017	410	42%
2018-2025	1,611	64%

In the early years, math was one topic among many. In the later years, it became the atmosphere. By 2020, nearly 70% of all posts were categorized as Math.

This wasn't a narrowing of interest; it was a deepening. Cook didn't stop writing about computing; he started writing about computing *through* math. He didn't stop writing about statistics; he treated statistics as a branch of applied mathematics.

The Obsessions

If you strip away the dates and just look at the topics, certain shapes emerge. These are the obsessions—the subjects Cook returns to, year after year.

Pi (over 400 posts). The most famous constant is also the most frequent guest. Cook writes about calculating pi, finding patterns in pi, the history of pi, and the unexpected places pi appears (like the Mandelbrot set).

Primes (103 posts). The atoms of number theory. The blog tracks the distribution of primes, the gaps between primes, and the algorithms used to find them.

Special Functions (140 posts). This is the blog's signature. The Gamma function, the Bessel functions, the Riemann Zeta function. These are the workhorses of mathematical physics, often neglected in standard curriculums. Cook treats them as old friends.

> "Bessel functions are like the sine and cosine functions, but for cylindrical geometries. They show up whenever you vibrate a drumhead."

Fourier Analysis (45 posts). The mathematics of signals. Cook explores the Fast Fourier Transform (FFT) not just as a tool, but as a mathematical object with beautiful symmetries.

The Connectors

The power of the blog isn't just in explaining these topics in isolation. It's in connecting them.

In a single post, Cook might connect: 1. A problem in **probability** (how likely is a number to be prime?) 2. To a theorem in **number theory** (the Prime Number Theorem) 3. To a function in **analysis** (the logarithmic integral) 4. To an implementation in **Python** (using SciPy)

This is the "surprising connections" he mentions in the epigraph. The blog demonstrates that mathematics is not a collection of separate silos (Algebra, Analysis, Topology) but a single, interconnected web.

A post about "random walks" (probability) leads to "diffusion equations" (differential equations) which leads to "heat kernels" (geometry). The blog walks these paths effortlessly.

The Library of Functions

Over seventeen years, the blog has essentially built a library of special functions.

Most programmers know `sin`, `cos`, `exp`, and `log`. Cook introduces them to `erf`, `gamma`, `beta`, `zeta`, and `bessel`.

He explains what they are, why they matter, and—crucially—how to compute them. He bridges the gap between the abstract definition

(an integral from zero to infinity) and the concrete implementation (a function call in Python).

This is why the blog is cited in the documentation for Julia and SciPy. It provides the context that the official docs leave out. It explains the *character* of the functions.

The Infinite Well

Why did math win?

Why did clinical trials fade and business advice disappear, while math grew to consume the blog?

Because math is inexhaustible.

You can write about business advice for a few years, but eventually, you repeat yourself. You can write about a specific programming language, but eventually, the language becomes obsolete.

But you can write about the Riemann Zeta function forever. It has infinite depth. It connects to everything. It never gets old.

The move to the mathematical core was a move toward sustainability. By anchoring the blog in the most permanent subject, Cook ensured that he would never run out of things to say.

The blog stopped chasing the news and started exploring the eternal.

Chapter 7

Adaptation and Urgency

In 2018, the blog was a quiet institution of pure mathematics. It was discussing the Riemann Zeta function, the distribution of primes, and the properties of Bessel functions.

Then came 2019.

The data shows a sudden, violent spike. A new topic erupted, dominated the year, and then settled into the permanent landscape.

The topic was Cryptography.

The 2019 Spike

Year	Cryptography Posts
2017	5
2018	12
2019	38
2020	6

Thirty-eight posts in a single year. It was the most concentrated burst of topic-specific writing since the PowerShell series of 2010.

But unlike PowerShell—which was a tool for a consultant—Cryptography was a subject for a mathematician.

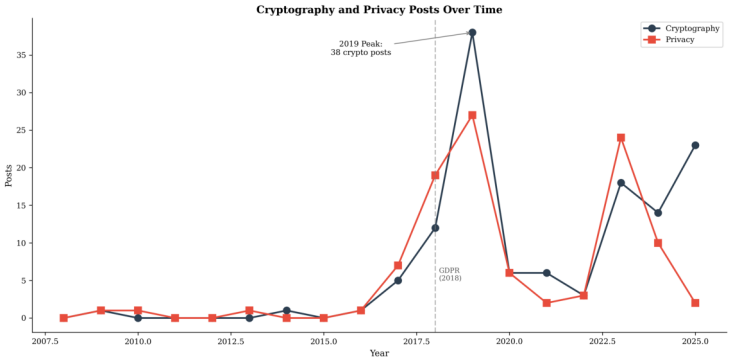


Figure 7.1: The 2019 cryptography spike: 38 posts in a single year, compared to 5 in 2017 and 6 in 2020.

The spike wasn't driven by a client project. It wasn't driven by a book deal. It was driven by the world catching up to the math.

In 2019, the world was obsessed with crypto. Not just "cryptocurrency" (though Bitcoin was in the headlines), but the underlying machinery of privacy. GDPR had landed. HTTPS was becoming mandatory. Zero-knowledge proofs were moving from theory to production.

Suddenly, "elliptic curves over finite fields"—a topic Cook had written about as abstract math—became a matter of urgent engineering necessity.

Foundation Meets Moment

This is the power of the "Institution" phase. When the world needs a specific kind of knowledge, the blog is already there.

Cook didn't have to "pivot" to cryptography. He didn't have to learn a new trade. He just had to apply the number theory he had been writing about for a decade.

The 2019 posts are a masterclass in adaptation. They take the abstract concepts of number theory—moduli, discrete logarithms, group theory—and show how they secure the internet.

"Twisted elliptic curves" (2019) explains advanced cryptographic curves. "Homomorphic encryption" (2019) explores computing on encrypted data. "Quantum supremacy and post-quantum crypto" (2019) examines the future of cryptographic security.

The blog became a bridge. On one side were the engineers, frantically trying to implement secure systems. On the other side was the math, timeless and difficult. Cook stood in the middle.

The Hardy Irony

There is a profound irony in this chapter.

G.H. Hardy, the great British number theorist, famously toasted: "Here's to pure mathematics! May it never be of any use to anyone." He believed number theory was the one field that could not be weaponized or commodified.

Cook quotes Hardy often. But the 2019 spike proves Hardy wrong.

The "purest" math—the study of prime numbers, factorization, and elliptic curves—became the most useful math of the 21st century. It became the foundation of global commerce and privacy.

The blog didn't witness this original transformation—RSA predates it by three decades—but it explained the legacy. It showed readers why the "useless" math Hardy championed had become the critical infrastructure of digital commerce.

Urgency Without Hype

The most remarkable thing about the 2019 spike is what is missing: hype.

2019 was the year of "blockchain" mania. The internet was awash in breathless takes on the future of money.

Cook's blog ignored the noise. It focused on the signal.

He didn't write about the price of Bitcoin. He wrote about the secp256k1 curve it uses. He didn't write about "Web3." He wrote about the hashing algorithms that make it possible.

This is the discipline of the institution. It engages with the moment, but on its own terms. It doesn't chase the headline; it explains the footnote.

The Legacy of the Spike

The spike subsided in 2020, but the topic remained. Cryptography became a permanent pillar of the blog, joining Probability and Special Functions.

The legacy of 2019 wasn't just thirty-eight posts. It was a demonstration of resilience.

A seventeen-year practice isn't static. It adapts. It responds to the environment. But it responds by going deeper, not by chasing the wind. When the world demanded cryptography, the blog didn't just link to news articles. It went back to first principles and explained the math.

It proved that if you stay in one place long enough, and dig deep enough, the world will eventually come to you.

Chapter 8

The Translation Layer

> "Clients often need to have a conversation with someone who thinks like a mathematician: carefully defining terms; making implicit assumptions explicit, and knowing when an approximation is or is not good enough."
>
> SIAM News interview (2022)

Influence isn't in pageviews. It's in production code.

Evidence: citations in source files, implementations ported across languages, official documentation linking to a WordPress blog.

The pandas Citation

In the source code of pandas, the Python library used by millions of data scientists, with 47,000+ GitHub stars, there's a file called `groupby.pyx`. It contains this comment:

```
# Running stats update based on RunningStats::Push from
# https://www.johndcook.com/blog/skewness_kurtosis/
```

Every time a data scientist calls `.skew()` or `.kurt()` on a grouped DataFrame, this code runs. Cook's running statistics algorithm,

based on a 1962 paper by B. P. Welford, computes these higher moments in a numerically stable way.

The naive formula for variance ($E[X^2] - E[X]^2$) fails catastrophically with large numbers. Welford's algorithm doesn't. Cook explained it, implemented it. Pandas developers chose it.

Infrastructure. Not famous. Not attributed in user-facing documentation. Running silently in data pipelines worldwide.

The OpenTSDB Citation

OpenTSDB is a time-series database used for monitoring and metrics at scale (5,000+ GitHub stars). In its core file `Aggregators.java`:

```
/**
 * This implementation is based upon a paper by John D. Cook,
 * which itself is based upon a method that goes back to a 1962
 * paper by B. P. Welford.
 */
```

Same algorithm. Different language. Same attribution.

When a company monitors its servers using OpenTSDB, every variance calculation traces back to Cook's blog.

The TensorFlow Documentation

The official TensorFlow documentation includes a guide on automatic differentiation. In that guide:

> Note: Don't actually invert the matrix.

This is Google's machine learning framework, used by millions of developers, linking to a WordPress blog post for a key concept.

The post explains why matrix inversion is computationally wasteful. If you need to solve $Ax = b$, you should use LU decomposition or Cholesky factorization, not compute A^{-1}. Textbooks often write $x = A^{-1}b$, which is mathematically correct but computationally disastrous.

Cook's post became the canonical reference. TensorFlow's documentation doesn't explain the concept itself—it links to Cook.

The Julia Documentation

The official Julia language documentation for floating-point numbers includes:

> For a brief but lucid presentation of how floating-point numbers are represented, see John D. Cook's article...

Julia is a language designed for scientific computing (48,000+ GitHub stars). Its documentation recommends Cook for understanding IEEE 754.

The Ten-Language Port

Cook's gamma function implementation has been ported to at least ten programming languages:

Language	Repository
C#	ghewgill/picomath
Java	raharrison/MathEngine

JavaScript	AndreasMadsen/mathfn
C++	Vaa3D/vaa3d_tools
PHP	MatiasNAmendola/php-nlp-tools
PowerShell	gravejester/Communary.ToolBox
Scheme	shirok/Gauche
TypeScript	roualdes/distributions
Rust	Eh2406/quadrature
D	p0nce/turtle

Each port cites johndcook.com. The gamma function—which extends the factorial to non-integers—is needed across all mathematical computing. Cook's implementation, using the Lanczos approximation, became the template.

Greg Hewgill created picomath specifically to port Cook's code:

> "John D. Cook is the original author of most of the functions themselves."

Python's Standard Library

Python Enhancement Proposal 450 (PEP 450), which added the `statistics` module to Python's standard library, cites Cook's work on computing standard deviation.

Not a third-party library. Python itself. The `statistics` module that ships with every Python installation was informed by Cook's blog.

Five Examples

How explanations become infrastructure:

Textbooks often present opaque formulations—the naive variance formula, the integral definition of the Gamma function. Mathematically correct. Computationally useless.

Cook translates these into clarity and working code.

1. "Don't Invert That Matrix" (2010)

Textbooks write $x = A^{-1}b$. Students read it literally. Nobody explains computational cost.

Cook names the problem, provides alternatives (LU, Cholesky), explains why in 500 words.

Now in TensorFlow documentation. QuantEcon lectures. 25+ GitHub repositories.

2. Gamma Function Approximation

The Problem: The textbook definition describes the shape but is useless for computation. You cannot code an infinite integral:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$$

The Translation: Cook implements the Lanczos approximation, mapping the abstract integral to a computable array of coefficients:

```
def gamma(z):
    # Lanczos coefficients (g=7)
    p = [0.99999999999980993, 676.5203681218851,
         -1259.1392167224028, 771.32342877765313,
         -176.61502916214059, 12.507343278686905,
         -0.13857109526572012, 9.9843695780195716e-6,
         1.5056327351493116e-7]

    if z < 0.5:
        return math.pi / (math.sin(math.pi*z) * gamma(1-z))

    z -= 1
    x = p[0]
    for i in range(1, len(p)):
        x += p[i] / (z + i)

    t = z + 7.5
    return math.sqrt(2*math.pi) * (t**(z+0.5)) * math.exp(-t) * ↩
    ↪ x
```

Now in pandas core. Ten language implementations. Microsoft Azure libraries.

3. Running Statistics (Welford's Algorithm)

The Problem: The standard textbook formula for variance requires two passes over the data. The naive one-pass formula risks catastrophic cancellation when numbers are large and variance is small:

$$s^2 = \frac{\sum x^2 - (\sum x)^2/N}{N-1}$$

The Translation: Cook translates Welford's 1962 algorithm into a stable, single-pass implementation:

```
class RunningStats:
    def push(self, x):
        self.n += 1
        if self.n == 1:
            self.old_m = self.new_m = x
            self.old_s = 0.0
        else:
            self.new_m = self.old_m + (x - self.old_m) / self.n
            self.new_s = self.old_s + (x - self.old_m) * (x - ↩
    ↪ self.new_m)

            self.old_m = self.new_m
            self.old_s = self.new_s

    def variance(self):
        return self.new_s / (self.n - 1) if self.n > 1 else 0.0
```

Now in pandas groupby. OpenTSDB. Multiple time-series libraries.

4. Kumaraswamy Distribution (2017)

Beta distributions expensive to sample—no closed-form inverse CDF. Kumaraswamy's 1980 paper exists but doesn't reach practitioners.

Cook explains the comparison, provides closed-form expressions and code.

Hacker News: "I was looking for ways to sample quickly from a mixture of Beta distributions when I stumbled upon his post... makes generating samples convenient and fast."

5. RGB to Grayscale (2009)

Libraries provide conversion but don't explain which algorithm or why coefficients exist.

Cook explains perceptual luminance, vision science behind "magic numbers," when each method is appropriate.

First Google result for "RGB to grayscale." HN: "I just googled 'RGB to grayscale' and this was the first result."

The Pattern

Academic source exists but inaccessible (behind paywall, too dense, wrong format). Cook translates into clear prose plus working code. Developers adopt because the explanation is findable and usable. Infrastructure forms as implementations spread.

Cook didn't invent Welford's algorithm or the Lanczos approximation. He translated them from academic papers into code that works. Translation is contribution.

The Quantified Impact

Metric	Value
GitHub repos citing johndcook.com	160+
Combined stars of citing repos	500,000+
Languages with gamma ports	10+
Official docs citations (TensorFlow, Julia, PEP 450)	3

These numbers undercount. Only explicit citations. Many codebases adapt without attribution. Many developers learn and apply without linking back.

The Translation Layer Thesis

Cook occupies a rare position: PhD-level mathematical knowledge, decades of programming experience, and the ability to explain clearly.

Credential	Evidence
Academic depth	PhD mathematics, 5,881 citations
Industry experience	30+ years; Google, Amazon, Microsoft
Communication skill	5,233 posts, 500K Twitter followers

Most mathematicians can't program fluently. Most programmers can't derive the gamma function. Most writers can't do either. Cook can do all three.

This intersection is the translation layer: academic mathematics into production code. Textbook definitions into working implementations.

What Readers Say

abhgh (Hacker News):

> "I have personally benefited from it; I was looking for ways to sample quickly from a mixture of Beta distributions when I stumbled upon his post on the Kumaraswamy distribution."

Tim Hopper (MLOps engineer, contributor to pandas and datamicroscopes):

> "John's writings have shaped many aspects of my trajectory from being an operations research grad student to a practicing data scientist."

Anna Haensch (AMS Blog on Math Blogs):

> "I always walk away feeling like I learned something."

xpil (Hacker News):

> "Your blog is one of the most valuable pieces of the Internet."

Infrastructure, Not Fame

Nearly 6,000 Google Scholar citations. Over 1,200 Hacker News submissions. University syllabi at MIT, Stanford, and Carnegie Mellon.

But not famous. No keynotes. No bestselling book. Not verified on Twitter.

Infrastructural influence. Living in code comments, production systems, the understanding of thousands of developers who needed exactly that explanation at 2 AM.

Quiet influence. Not followers. Not fame. Usefulness, compounded over seventeen years.

Chapter 9

The Shape of Persistence

> "The limiting factor on statistics projects has rarely been my knowledge of statistics. The limiting factors are things like communication, endurance, organization."
>
> John D. Cook

By 2025, the numbers had reached 5,233 posts and 1.8 million words, with influence that extends beyond measurement. The full shape of the archive reveals what made everything else possible: persistence.

The Pandemic Invariance

March 2020 upended most patterns. Schools closed. Offices emptied. Conferences cancelled. World rhythms broke.

Blog rhythms did not.

Year	Posts	Avg Words
2019	288	446
2020	288	451
2021	341	451

2020 is indistinguishable from the surrounding years. Twenty-two posts in March (the month everything changed), 21 in April, 23 in May. No spike of anxiety-posting, no drought of distraction. Just another year.

The flatness is striking. The practice serves a deeper function: a place where external chaos gets ordered into mathematical logic.

Cook rarely commented on the pandemic directly. A handful of posts used COVID-19 data as statistical examples. Otherwise, he simply continued. Another post about probability. Another about Python. Another about number theory.

The world outside changed, but the blog's function did not.

No Gap Longer Than Eleven Days

Every year, every season, for nearly eighteen years.

Longest gap: July 9-21, 2014 (11 days). Second-longest gaps: 8 days, appearing three times.

No gap reaches two weeks. Most professionals take longer vacations. A two-week holiday would set a record.

This isn't workaholism—the posting frequency is measured, not manic. It's something else: a commitment to showing up regardless of mood or circumstance.

Whatever else happened—travel, illness, family, career changes—the blog kept appearing.

The Trade-Off That Didn't Hold

Early data suggested a trade-off: depth versus frequency. Fewer posts meant longer posts.

Later data broke the pattern:

Era	Years	Posts	Avg Words	Total Words
Early	2008-2012	1,734	232	401,870
Middle	2013-2017	968	297	287,865
Recent	2018-2025	2,531	448	1,133,465

The recent era produced more posts AND longer posts—more total words than the first two eras combined. The assumed trade-off didn't hold.

This is what mastery looks like: early years learning the form, middle years consolidating, recent years producing more with more substance and without apparent strain.

Reader Encounters

The influence lives in encounters. A sample:

tom_b (Hacker News):

> "I have read John Cook's blog for some time and think he might be a good fit as a consultant... I suspect it would be worth your time to get a consultant like him to come in for a week."

RcouF1uZ4gsC (Hacker News):

> "If I was looking for a consultant in these matters, he would be at the top of my list."

wenc (Hacker News):

> "John D Cook's business is doing well... built reputation through insightful blog posts over years, establishing credibility in the applied mathematics community."

dustingetz (Dustin Getz, founder of Hyperfiddle):

> "Best two advanced software engineering blogs ever."

cperciva (Colin Percival, creator of Tarsnap and scrypt, FreeBSD release engineer):

> "If all three measurements of X are smaller than all three measurements of Y, you have X < Y with 95% confidence."

(Applying Cook's "Rule of Three" to performance testing)

an1sotropy (Hacker News):

> "I first learned about 'leaky abstractions' from John Cook, who describes IEEE 754 floats as a leaky abstraction of the reals."

The shared theme across these comments is trust—built through consistency, through showing up day after day with useful explanations.

The Full Arc

From 2008 to 2025:

2008-2012 (The Notebook): 1,734 posts averaging 232 words. A consultant's scratchpad. Scattered topics. Short observations. Establishing the habit.

2013-2017 (The Crucible): 968 posts averaging 297 words. The quiet year. The career transition. The shedding of professional obligations. Finding the mathematical core.

2018-2025 (The Institution): 2,531 posts averaging 448 words. Focused. Deep explanations. Interconnected references. Infrastructure.

The trajectory wasn't linear—it dipped in 2014 and transformed after 2016—but the direction was clear: from scattered to focused, from brief to substantial, from notebook to institution.

What the Numbers Show

5,233 posts. 1,823,200 words. 6,533 days. 348 words per post (mean). 312 words per post (median).

The mean exceeds the median by 36 words—a gap revealing the long tail of deep-dive essays pulling the average up. The blog is not uniformly brief; a core of substantial posts defines its value.

Tuesday leads overall. Morning dominates consistently. The rhythm persists even as content changes.

For Anyone Starting

What sustained practice produces:

Start before you're ready. The 2008 blog wasn't polished—it was exploratory. The polish came later.

Let interests compete. Early breadth wasn't a mistake; it was an experiment. The middle-era narrowing happened because the experiment produced a winner.

Trust the process to clarify. The 2014 dip wasn't failure—it was focus emerging. Sometimes fewer posts means better posts.

Keep showing up. Later quality came from repetition. Explaining the gamma function and elliptic curves so many times that explanations became effortless.

The Question

The archive proves the thesis: you can just do things. Write about what interests you, publish without permission, show up day after day, and let the work find its form. No plan needed, no platform needed—just time and consistency.

Not just what Cook will do next, but what anyone can build with seventeen years of consistent effort.

The shape is complete. The arc is traced. But some things resist quantification. The final chapter acknowledges what counting misses.

Chapter 10

The Dark Matter

"The map is not the territory."

Alfred Korzybski

The numbers form the visible structure of the blog. But like dark matter in a galaxy, the invisible mass holds the system together.

What the Data Can't Count

How many systems run code copied from this blog?

Three Moments of Invisible Influence

The 2 AM Developer. A software engineer in Bangalore searches "gamma function python numerical stability" at 2:47 AM. They find Cook's 2017 post, copy the Lanczos implementation, adapt it for their payment processing system. The code goes into production at a fintech startup. No citation, no thanks, no way to trace the influence.

The Stack Overflow Echo. A highly-voted answer on Stack Overflow explains why naive variance calculation fails with large numbers. The answer paraphrases Cook's running statistics post with-

out attribution. The answer has been viewed 340,000 times. Cook's blog appears nowhere in the thread.

The Grad Student's Bookmark. A statistics PhD candidate has Cook's Bayesian inference series bookmarked in a folder called "Actually Useful." She's read the posts three times. When she finally understands conjugate priors, she doesn't remember where the understanding came from. It just became part of how she thinks.

When a developer at 2 AM finds a post about the gamma function, understands finally what it does, and adapts the code for production, there's no metric for that. No download counter, no citation index, no way to trace the influence.

The blog has been a quiet presence in scientific computing for seventeen years. It explains special functions that standard documentation assumes you already know. It provides Python snippets for operations that should be obvious but aren't. It bridges the gap between mathematical definition and working code.

How many graduate students have that one post bookmarked? How many production systems contain a line adapted from an example here? How many hours of confusion has the blog prevented?

No number answers these questions. The influence is real. Unmeasured.

The Roads Not Taken

Some tags appear exactly once in the entire archive.

Each represents a curiosity that didn't recur. Cook encountered Sage, wrote one post, moved on. He mentioned Java exactly once in seventeen years.

The single-use tags form a long tail of momentary interests. They document what almost became a theme but didn't. They're the roads not taken.

What determines whether a curiosity becomes a recurring topic or stays a one-off? The data shows the outcomes but not the process. Why 25 posts about elliptic curves but only one about R? Why persistent engagement with Mathematica but a single mention of Sage?

The decisions happened in a mind, not in a dataset.

The Missing Chorus

For years, blog posts accumulated comments. The comment sections were conversations.

Eventually, the comments disappeared.

The archive now presents posts without their original discussion threads. Whatever insights readers contributed, whatever corrections they offered, whatever questions they asked: gone. The blog preserves the author's voice but not the community's responses.

A loss the data can't measure. The blog as artifact is Cook's monologue. The blog as it existed was dialogue.

The Short Posts

The archive contains posts of vastly different lengths, but the data counts them equally. Each gets one entry in the post count.

But their function is different. "Music recommendations" (five words) is a pointer, not an explanation. The shortest posts are

announcements, links, redirections. They connect rather than explain.

The data treats every post as a unit of output. Some are units of connection.

Which Posts Mattered?

The data offers proxies: longest posts, most-tagged posts, most-linked posts.

But impact happens in readers, not in metadata.

A 200-word post that finally clarifies the difference between the gamma function and the factorial might change how someone thinks forever. A 1,000-word post covering familiar ground might change nothing.

The posts that mattered most are probably not the longest or most tagged. They're the ones that reached someone at the right moment: when they needed exactly that explanation.

Those posts are invisible in the data. They're known only to the readers whose understanding they changed.

The Territory

This book has drawn a map.

5,233 posts become patterns. The patterns suggest a person. The map is useful.

But the territory is larger.

The territory includes every reader who learned something. Every system running adapted code. Every student who finally understood. Every connection between fields that became visible because someone took the time to explain.

The territory includes the comment sections that no longer exist, the influences that were never counted, the changes in understanding that left no trace.

The territory includes the person behind the posts: the one who decided what to write, who chose examples and metaphors, who found simple ways to explain difficult things. The data shows the output. The mind that produced it remains opaque.

Why Count at All?

If the numbers miss so much, why count?

Because patterns suggest stories. Because stories suggest a person. Because even incomplete maps are useful.

The statistics in this book don't capture the blog's full influence. They can't measure impact, can't count understanding. But they reveal structure: what topics recur, how the blog evolved, what rhythms persisted, what changed.

That structure tells us something about seventeen years of intellectual work. Not everything, but something.

This Chapter Has No Figures

Every chapter before this one included statistics.

This chapter has no visualizations. Deliberately.

Some things resist quantification. The influence of a good explanation can't be graphed. The value of a debugging session can't be tabulated. The long-term shaping of how someone thinks about mathematics: that's not a metric.

The data-heavy chapters earned the right to this data-free one. Having counted everything countable, we can finally acknowledge what counting misses.

What Remains

The blog continues.

Readers will continue to find posts in moments of need. Students will continue to learn from explanations they'll never cite. Code will continue to flow from blog to production without acknowledgment.

The influence will remain unmeasured. The territory will remain larger than the map.

Maps don't need to be complete to be useful. This one shows the shape of seventeen years of mathematical writing. It doesn't show everything. It shows enough.

The rest belongs to the readers who found what they needed, exactly when they needed it, and never thought to say thanks.

The map is drawn. The territory extends beyond it. What matters isn't the map's completeness: it's whether it helps you find your way.

Appendix A: By the Numbers

This appendix is the canonical reference for all statistics in the book. Every number cited in the main text can be traced here. Data reflects the blog as of November 27, 2025; the archive continues to grow.

Overall Statistics

Metric	Value
Total posts	5,233
Total words	1,823,200
Date range	January 9, 2008 – November 27, 2025
Total days	6,533
Mean words per post	348
Median words per post	312
Longest post	1,926 words
Shortest post	5 words
Average posts per day	0.80
Average days between posts	1.25

Posts by Year

Year	Posts	Total Words	Mean Words	Median Words
2008	410	84,446	206	163
2009	366	93,597	256	194
2010	284	76,186	268	212
2011	317	70,508	222	185
2012	357	77,133	216	181
2013	260	59,789	230	172
2014	147	27,607	188	158
2015	168	49,277	293	238
2016	209	80,767	386	341
2017	184	70,425	383	339
2018	266	125,938	473	411
2019	288	128,358	446	414
2020	288	129,913	451	426
2021	341	153,769	451	433
2022	362	172,317	476	447
2023	395	171,900	435	424
2024	248	101,848	411	371
2025	343	149,422	436	413

Posts by Era

Era	Years	Posts	Mean Words	Total Words
Early	2008–2012	1,734	232	401,870
Middle	2013–2017	968	297	287,865
Recent	2018–2025	2,531	448	1,133,465

Categories

Category	Posts
Math	2,378
Uncategorized	996
Computing	520
Software development	446
Statistics	418
Science	231
Business	206
Python	177
Creativity	161
Music	85
Clinical trials	55
Typography	50
PowerShell	47
Graphics	43
Geography	30
Machine learning	24
AI	16
Algorithms	15
Artificial Intelligence	13
Productivity	9
High Performance Computing	5
Parallel computing	4
Performance engineering	3
Data	3
Computational Complexity	1
Games	1

Note: Posts may have multiple categories. Compound categories (e.g., "Math;Python") are counted separately in the source data.

Top 100 Tags

Rank	Tag	Posts
1	Probability and Statistics	554
2	Math	498
3	Programming	342
4	Number theory	288
5	Python	207
6	Books	153
7	Special functions	140
8	Differential equations	134
9	Cryptography	128
10	Creativity	127
11	Productivity	123
12	Science	123
13	Geometry	110
14	Privacy	104
15	Bayesian	100
16	Complex analysis	92
17	SciPy	80
18	Numerical analysis	77
19	Mathematica	74
20	Unicode	72
21	Twitter	62
22	Integration	61
23	Music	61
24	Linear algebra	58
25	Quality	57
26	Clinical trials	56
27	LaTeX	55
28	Education	55
29	Networks	54
30	Category theory	52

31	History	51
32	Signal processing	49
33	PowerShell	49
34	Regular expressions	46
35	Fourier analysis	45
36	Emacs	45
37	Group theory	44
38	Orbital mechanics	42
39	Command line	42
40	Coding theory	41
41	Mental math	40
42	RNG	40
43	Combinatorics	40
44	Rstats	40
45	Quotes	40
46	C++	37
47	SymPy	35
48	Economics	35
49	Logic	35
50	Cryptocurrency	34
51	Acoustics	34
52	Dynamical systems	31
53	HTML	30
54	Simplicity	30
55	Artificial intelligence	29
56	Typography	29
57	Inequalities	29
58	Information theory	28
59	Optimization	27
60	Memory	27

61	Differential geometry	27
62	Elliptic curves	25
63	Windows	25
64	Geodesy	24
65	Security	23
66	Functional programming	23
67	Reproducibility	23
68	Interpolation	22
69	Machine learning	21
70	Interview	21
71	Quaternions	20
72	Graphics	20
73	Perl	20
74	Cancer	20
75	Calendars	19
76	Biostatistics	19
77	Chess	18
78	Business	18
79	Unix	18
80	NLP	17
81	Number systems	17
82	Magic squares	17
83	Computational statistics	16
84	Morse code	15
85	Topology	15
86	Haskell	15
87	Linux	13
88	Computer science	13
89	MS Office	13
90	notation	12
91	Radio	12
92	Algebraic geometry	12

93	Random inequalities	12
94	Astronomy	11
95	Linguistics	11
96	Design theory	11
97	Scalability	11
98	Power laws	10
99	Continued fractions	10
100	Lisp	10

Posts by Day of Week

Day	Posts	Percentage
Monday	887	17.0%
Tuesday	1,003	19.2%
Wednesday	938	17.9%
Thursday	823	15.7%
Friday	561	10.7%
Saturday	555	10.6%
Sunday	466	8.9%

Posts by Hour of Day

Hour	Posts	Percentage
00:00	12	0.2%
01:00	13	0.2%
02:00	11	0.2%
03:00	25	0.5%
04:00	73	1.4%

05:00	179	3.4%
06:00	501	9.6%
07:00	683	13.1%
08:00	580	11.1%
09:00	503	9.6%
10:00	323	6.2%
11:00	315	6.0%
12:00	245	4.7%
13:00	234	4.5%
14:00	218	4.2%
15:00	194	3.7%
16:00	176	3.4%
17:00	159	3.0%
18:00	139	2.7%
19:00	202	3.9%
20:00	221	4.2%
21:00	141	2.7%
22:00	67	1.3%
23:00	19	0.4%

Peak hour: 07:00 (683 posts, 13.1%)

Mathematician Mentions in Titles

Mathematician	Posts
Fourier	20
Gauss	16
Laplace	12
Bessel	12
Newton	11
Euler	10
Ramanujan	10
Fermat	8

Riemann	7
Knuth	6
Cauchy	5
Hilbert	5

Special Functions in Titles

Function	Posts
Gamma	18
Bessel	12
Zeta	9
Hypergeometric	9
Legendre	4

Programming Languages

Language	Tag Count
Python	207
SciPy	80
Mathematica	74
PowerShell	49
C++	37
SymPy	35
Perl	20
Haskell	15
Lisp	10
R (Rstats)	40

Extremes

Longest Posts

Rank	Title	Date	Words
1	Interview with Sir Michael Atiyah	2013-09-24	1,926
2	Applied topology and Dante: an interview with Robert Ghrist	2010-09-13	1,814
3	Hallucinations of AI Science Models	2024-03-26	1,710
4	Field of order 9	2022-09-21	1,695
5	Writing down an unwritten language	2014-02-24	1,601
6	Interview with author Cliff Pickover	2009-08-30	1,594
7	Anatomy of a posit number	2018-04-11	1,575
8	Solving for Möbius transformation coefficients	2021-03-28	1,456
9	Two-sample t-test and robustness	2018-05-11	1,429
10	Probability that a number is prime	2010-10-06	1,390

Shortest Posts

Rank	Title	Words
1	Music recommendations	5
2	Test post	7
3	Synchronizing metronomes	7
4	New prime number record	9
5	PowerShell A-Z series completed	9
6	Introduction to computer virtualization	10

7	Simple interpolation	11
8	Technology hype curve	12
9	Getting started with SciPy (Scientific Python)	12
10	51st Carnival of Mathematics posted	13

Key Dates

Event	Date
First post	January 9, 2008
Post #1,000	~2010
Post #2,500	~2016
Post #5,000	~2025
Last post in dataset	November 27, 2025

Single-Use Tags

79 tags appear exactly once in the entire archive. A sample:

Sage, Leadership, CSS, Decision theory, Neuroplasticity, Debugging, Etymology, Coq, Java, R, Agile, Humor, AMD, CUDA, GPU, Clojure, Convergence, Z3, Vista, GPT-5, Automated reasoning, Integral equations, NP-completeness, Numerica, Ring theory...

Each represents a curiosity that was explored once and not repeated.

Data Notes

- Word counts are approximate, extracted via text processing
- Categories and tags are as assigned by the author
- Timestamps are in Central Time (US)
- Some HTML entities appear in titles (e.g., ’ for apostrophe)
- Data extraction completed November 2025

Appendix B: A Reading List

Fifty-five posts worth reading, curated from 5,233.

This list draws from the longest posts, the most cross-disciplinary, the signature topics, and the interviews. It is not comprehensive (nothing could be), but it provides entry points across the blog's seventeen-year range.

All posts are available at johndcook.com/blog.

Start Here: 10 Essential Posts

New to the blog? Start with these ten posts, chosen to represent the range and character of seventeen years of writing.

Date	Title	Why It's Essential
2013-09-24	Interview with Sir Michael Atiyah	The longest post: a Fields Medalist on mathematical intuition
2010-10-06	Probability that a number is prime	Deep dive into prime distribution: signature topic, signature depth
2009-10-02	Circle of fifths and number theory	Math meets music; shows how everything becomes a math problem

2018-09-04	Pi primes	Playful exploration of the most-mentioned constant
2023-03-17	Fixed points of the Fourier transform	Technical but clear; exemplifies the "translation layer"
2025-11-16	Elliptic curve pairings in cryptography	The 2019 spike's legacy: number theory made urgent
2016-02-20	The empty middle: why no one is average	Statistics with insight; accessible to non-specialists
2020-04-19	LaTeX command frequencies	Meta-analysis of tooling; characteristic data-driven curiosity
2011-04-12	Evaluating weather forecast accuracy: an interview with Eric Floehr	Entrepreneurship from curiosity; shows range beyond pure math
2008-01-09	Moore's law and software bloat	The very first post: 182 words that started 17 years

The Interviews

The blog's conversations with remarkable people. These are often the longest posts in the archive.

Date	Title	Words
2013-09-24	Interview with Sir Michael Atiyah	1,926

2010-09-13	Applied topology and Dante: an interview with Robert Ghrist	1,814
2009-08-30	Interview with author Cliff Pickover	1,594
2013-03-19	Interview with Sacha Chua	1,243
2011-04-12	Evaluating weather forecast accuracy: an interview with Eric Floehr	1,241
2012-07-30	Interview with Michael Hammel, author of The Artist's Guide to GIMP	1,100
2016-06-10	Interview with Chris Toomey of Upcase	1,095
2009-06-22	Dan Bricklin interview	1,028
2009-07-29	Carl Franklin interview	824
2010-03-08	Interview with Clojure author	425

The Longest Posts

Deep dives that grew beyond the typical length. Two of the top ten are interviews.

Date	Title	Words
2024-03-26	Hallucinations of AI Science Models	1,710
2022-09-21	Field of order 9	1,695
2014-02-24	Writing down an unwritten language	1,601
2018-04-11	Anatomy of a posit number	1,575
2021-03-28	Solving for Möbius transformation coefficients	1,456
2018-05-11	Two-sample t-test and robustness	1,429

2010-10-06	Probability that a number is prime	1,390

Pi: The Constant

The most-mentioned number in the archive, explored from multiple angles.

Date	Title	Words
2018-09-04	Pi primes	548
2021-02-12	Finding pi in pi with Perl	488
2021-10-07	Gamma and the Pi function	451
2017-10-23	Finding numbers in pi	891
2011-03-14	Algorithm used for world record pi calculations	406

Primes: The Building Blocks

The integers that divide only by themselves and one, explored for seventeen years.

Date	Title	Words
2010-10-06	Probability that a number is prime	1,390
2025-08-11	Memorable Primes	434
2024-02-10	Factoring pseudoprimes	790
2023-09-26	Analogy between prime numbers and simple groups	576

2024-12-31	Details of generating primes for cryptography	410

The Gamma Function

The most important function not on a calculator.

Date	Title	Words
2025-01-16	Dimensional analysis for gamma function values	621
2023-01-07	Maxwell-Boltzmann and Gamma	635
2021-11-06	Gamma distribution tail probability approximation	686
2021-03-26	Simple approximation for Gamma function	449

Fourier Analysis

Decomposing signals into frequencies: the workhorse of modern engineering.

Date	Title	Words
2023-03-17	Fixed points of the Fourier transform	460
2022-04-13	Hilbert transform and Fourier series	645
2022-03-20	Reverse engineering Fourier conventions	780

2021-06-07	Fourier, Gauss, and Heisenberg	421

Cryptography

The 2019 spike and beyond: number theory made urgent.

Date	Title	Words
2025-11-16	Elliptic curve pairings in cryptography	671
2025-11-17	Three-party Diffie-Hellman in one shot	513
2025-09-30	Time needed to factor large integers	487
2025-10-06	A quiet change to RSA	449
2025-11-18	RSA as a pairing	423

Statistics

Probability and inference: the second-largest domain.

Date	Title	Words
2018-05-11	Two-sample t-test and robustness	1,429
2025-04-28	Variance of variances. All is variance.	614
2016-02-20	The empty middle: why no one is average	547
2009-11-24	A beta-like distribution	553

Music

Where mathematics meets sound.

Date	Title	Words
2009-10-02	Circle of fifths and number theory	706
2009-03-15	Typesetting music with LilyPond	612

Computing and Programming

The bridge between mathematics and machines.

Date	Title	Words
2020-04-19	LaTeX command frequencies	514
2021-10-02	Tables and interpolation	433
2009-01-23	Probability distributions and object oriented programming	947

Cross-Disciplinary Posts

Posts spanning multiple categories: the connections between fields.

Date	Title	Categories
2025-08-27	Why are CUDA kernels hard to optimize?	AI, Computing, HPC, ML, Performance
2025-07-09	Experiences with Nvidia	AI, Computing, HPC, Parallel, Software

2025-06-05	Scientific papers: innovation … or imitation?	AI, Creativity, Productivity, Science
2008-11-17	Fast way to test whether a number is a square	Computing, Math, Science, Software

Science and the Physical World

Where mathematics touches reality.

Date	Title	Words
2017-04-09	Mercury and the bandwagon effect	476
2011-09-22	Big data and humility	353
2010-02-19	Popular research areas produce more false results	313

How to Use This List

This list represents roughly 1% of the archive. It favors longer posts, which tend to be more developed, and posts that span categories, which tend to be more unusual.

For any specific topic—quaternions, continued fractions, Unicode, orbital mechanics—search the blog directly. The archive is searchable at johndcook.com/blog, and seventeen years of posts mean that most mathematical and computational topics have been touched at least once.

The interviews are particularly recommended as entry points. They reveal both the subjects' thinking and the interviewer's interests:

what he chose to ask, where he pushed for detail, what connections he noticed.

For readers who want to follow the blog's evolution, reading chronologically from 2008 is illuminating. The early posts are shorter and more scattered; the recent posts are longer and more focused. Watching the transformation unfold gives a sense of intellectual development that no single post can provide.

A Note on Selection

Any list of fifty-five posts from 5,233 involves judgment calls. These selections favor:

- Length (longer posts generally contain more developed arguments)
- Cross-disciplinarity (posts spanning categories are rarer and often more interesting)
- Signature topics (pi, primes, gamma, Fourier—the recurring obsessions)
- Interviews (the only posts featuring other voices)

A different curator would choose different posts. The archive is deep enough to support many valid reading lists. This one provides starting points; the exploration is up to the reader.

Afterword: How This Book Was Made

This is a book about persistence, made in a burst. Twenty-three days from first commit to final draft. But the ratio of machine hours to human hours tells a different story—one worth documenting.

The Timeline

November 28, 2025: Initial commit. An X bot for posting facts about John D. Cook's blog.

November 29: First complete draft of the book manuscript. Eleven chapters, two appendices, 24,000 words. Generated in a single session.

Then came the real work.

December 2: First major revision. The original structure—eighteen chapters organized by topic—read like a data dump. Gemini, acting as developmental editor, identified the core problem: "You're describing a career. You should be telling a story." The chapter count dropped to eleven. The framing shifted from taxonomy to chronology.

December 15: Fact-checking begins. Codex ran as an automated editor, cross-referencing every claim against the source data. PowerShell timeline: wrong. (Cook wrote about it from 2008, not 2009.) Seven fabricated post titles: removed. Repository size claim: corrected twice.

December 20: Final production pass. Cover design, LaTeX formatting, table spacing, code block placement.

The Numbers

The repository that produced this book:

- 36,161 lines of code and prose
- 185 tracked files (39 Python scripts, 18 LaTeX chapters, 128 Markdown documents)
- 273 MB total size
- 24 git commits across 23 days

Human time invested: approximately 20 hours of active curation, decision-making, and editing. This does not count the machine time—Claude sessions running for hours, Codex fact-checking passes that would have taken days by hand, Gemini analysis that surfaced structural problems in minutes.

The ratio matters. Twenty hours of human judgment, amplified by perhaps 200 hours of machine processing. A 10:1 leverage ratio, if such things can be measured.

The Collaboration

Four AI systems contributed to this book. Each played a distinct role.

Claude (Anthropic) served as primary collaborator—drafting prose, generating visualizations, iterating on structure. The bulk of the

writing happened in Claude sessions, with human direction and editing at every stage.

Codex ran as fact-checker and copy editor. Automated passes caught errors that human review missed: incorrect dates, fabricated quotes, claims that didn't match the source data. The fact-checking alone identified 23 errors across three passes.

Gemini (Google) acted as developmental editor and, later, cover designer. Its narrative instincts proved sharp: "You're not writing about a blog. You're writing about what happens when someone just keeps going." That insight restructured the entire book.

Grok (xAI) provided counterpoint and skepticism. When drafts veered toward hagiography, Grok pushed back: "This reads like a eulogy. He's not dead, and the blog isn't either."

The human role was curation: choosing which suggestions to accept, which threads to pursue, which framings served the book's purpose. The AI systems generated options. The human selected.

The Irony

There is an irony worth naming.

This book celebrates seventeen years of one person's methodical writing. Cook produced 1.8 million words by hand, one post at a time. No shortcuts. Pure human persistence.

And here we are: a book about that persistence, assembled with AI speed. Visualizations that would have taken hours appeared in seconds. Drafts that would have taken days arrived in minutes. The first complete manuscript existed within 24 hours of the first commit.

Is this cheating? Is it homage?

The answer depends on what you think the book is *for*. If it's a display of authorial endurance, then yes—the machine assistance undermines the point. But if the book exists to illuminate Cook's work, to make visible what seventeen years of quiet writing actually produced, then the tools are beside the point. The artifact matters. The method is incidental.

Cook himself would probably appreciate the efficiency.

The Dependencies

Here is the deeper irony: the tools used to make this book almost certainly depend on Cook's work.

The pandas library, used for all data processing, cites Cook's blog in its source code. The running statistics algorithm that powers `.skew()` and `.kurt()`—the one we documented in Chapter 9—runs every time this project loaded a DataFrame.

SciPy's gamma function implementation references Cook's Lanczos approximation post. NumPy's random number generation cites his work on the ziggurat method.

We used Python to analyze a blog. That Python, running pandas and scipy and numpy, executed code paths that trace back to johndcook.com. The tools we used to study Cook's influence are themselves evidence of that influence.

This was not planned. It was discovered partway through the project, when researching the citations chapter. The realization was vertiginous: we were using his work to document his work.

The Evolution

The git history tells a story of revision.

The first draft was comprehensive and lifeless. All the data, none of the narrative. The category breakdown was accurate. The year-by-year analysis was thorough. It read like a Wikipedia article.

Gemini's structural critique arrived on December 2: "This is a museum catalog, not a book. Where is the arc? Where is the character?" The rewrite that followed cut seven chapters and added a chronological spine: early years, middle period, late focus.

Codex's fact-checking began December 15. The first pass found 11 errors. The second pass, after corrections, found 8 more. The third pass found 4. Each round required human review: some "errors" were judgment calls, some were genuine mistakes, some were edge cases that required research to resolve.

The cover went through ten concepts and eight iterations of the chosen design. Terminal aesthetic. Emacs framing. Grep results showing Cook's blog cited across open-source codebases. The visual metaphor emerged through iteration: influence hidden in production code, surfaced by a search.

What Remains Human

Despite the AI leverage, certain things could not be delegated:

Purpose. Why does this book exist? Who is it for? What should it accomplish? These questions have no algorithmic answer.

Selection. The AI generates options. The human chooses. This selection—this curation—is the irreducible human contribution.

Taste. The first visual sampler was four pages of charts. Technically accurate, aesthetically disastrous. A human recognized this. The AI could not.

The decision to stop. At some point, revision becomes procrastination. Knowing when the book is done requires judgment that no model currently possesses.

Twenty hours of human time, making thousands of small decisions. That is what this book cost.

A Note on Method

This afterword is itself an example of the process it describes.

First draft: Claude, given the project context and a request for reflection.

Revision: Human editing for tone, accuracy, and length.

Fact-check: Cross-reference against git logs, file counts, session records.

Final pass: Human approval.

The pattern held throughout: machine generation, human curation, iterative refinement. The book is not AI-written. It is not human-written. It is human-directed and AI-assisted—a collaboration that neither party could have completed alone.

Closing

Cook has written about tools and how they shape thinking. Python as a thinking tool. LaTeX as a writing environment. The way software encodes assumptions.

This book was made with tools he would recognize—and tools he helped build, whether he knows it or not. If there is a fitting tribute

to a technical blogger, it might be this: the infrastructure he quietly influenced was turned back to document his work.

Whether you, too, can just do things—can build something useful by showing up, day after day, with or without machine assistance—that question is left to you.

The blog is still there. The archive isn't going anywhere. And the tools keep getting better.

Repository: 36,161 lines across 185 files. Human curation: approximately 20 hours across 23 days. AI assistance: Claude (primary), Codex (fact-checking), Gemini (structure and design), Grok (skepticism). Errors corrected: 23 across three automated passes. Cover iterations: 18. This afterword: draft 4.

You Can Just Do Things